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Paleomagnetism of ca. 1.35 Ga sills in northern North China Craton and implications for paleogeographic reconstruction of the Mesoproterozoic supercontinent

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ABSTRACT

We report paleomagnetic data on precisely dated Mesoproterozoic sills, in zircon and baddeleyite $^{207}Pb/^{206}Pb$ ages of 1345 ± 12 and 1353 ± 14 Ma respectively, intruding the Xiamaling and Wumishan formations in North China with an aim of evaluating how the North China Craton (NCC) was involved in the evolution of the Columbia supercontinent from ca. 1.78 to 1.35 Ga. After systematic thermal demagnetization we isolate a high temperature characteristic remanence from 18 sites with a tilt-corrected site-mean direction of $D = 294.4^\circ$, $I = -31.7^\circ$ and $\alpha_{95} = 4.3^\circ$, corresponding to a mean paleomagnetic pole at $\lambda = 5.9^\circ$N, $\varphi = 359.6^\circ$E ($N = 18$) with angular standard deviation of 10.0° and $A_{95} = 4.3^\circ$. Rock magnetic experiments, light microscopy and scanning electron microscopy all indicate that the main magnetic minerals in the sampled sill rocks are medium sized titanomagnetites. The positive fold test and fresh titanomagnetite grains suggest that the characteristic remanence is likely to be a primary acquired at ca. 1.35 Ga. The preliminary paleogeography reconstruction based upon the well-matched apparent polar wander paths between the Laurentia–Baltica–Siberia united block and the NCC supports the interpretation that the NCC may have drifted together with this united block before ca. 1.35 Ga during the late history of the Columbia supercontinent. We argue that the NCC was located in low latitudes and kept tight connections with the Siberian and Indian cratons during the interval between ca. 1.8 and 1.35 Ga, and that the ca. 1.35 Ga sills widely developed in the Mesoproterozoic Xiamaling and Wumishan formations, as well as the ca. 1.3–1.2 Ga magma events well-developed in Siberia and northern margin of the NCC, may represent the complete fragmentation of the NCC with the neighboring cratons.

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1. Introduction

The Paleo-Mesoproterozoic supercontinent Columbia (Nuna) has been discussed for more than a decade and is considered to have been an amalgamation of all cratons of the globe (Zhao et al., 2003a; G. Zhao et al., 2004). At present, it is widely considered that the Columbia supercontinent was amalgamated at ca. 2.0–1.8 Ga (Zhao et al., 2003a; Hawkesworth et al., 2009), and broke up at ca. 1.5–1.3 Ga (G. Zhao et al., 2004; Hou et al., 2008a). However, the evolvement of its configuration has been hotly debated and remains poorly constrained. Hitherto, there have been at least three main different models based on the distribution of ca. 2.1–1.8 Ga orogens (Condie, 2002; Rogers and Santosh, 2002; Zhao et al., 2003a; G. Zhao et al., 2004) with recent studies presenting some interesting suggestions on the basis of paleomagnetic data and geological correlations (Hou et al., 2008a; Evans and Mitchell, 2011). In a word, no consensus could be reached so far for the paleogeographic configuration of a possible Paleo-Mesoproterozoic supercontinent Columbia because of both geological complexities and paucity of high quality paleomagnetic data. The latter has been widely accepted to be the only a quantitative constraint on the paleogeographic reconstruction of supercontinent and the relationship of the North China Craton (NCC, Fig. 1) to neighboring cratons is still an enigma due to the lack of precise age-controlled high quality paleomagnetic data. Wilde et al. (2002) proposed that the NCC might be connected with the Baltica Craton in the light of lithologic correlation and the distribution of major crustal components, whilst Condie (2002) argued that it should be adjacent to the Siberian Craton, and Zhao et al. (2003b) regarded the Indian Craton as the NCC's neighbor. On the other hand, Wu et al. (2005) and Pei et al. (2006) reported respectively two ca. 1.35 Ga paleomagnetic poles for the NCC based on an $^{40}Ar–^{39}Ar$ dating age of the Yangzhuang Formation (Wang et al., 1995). Although these two

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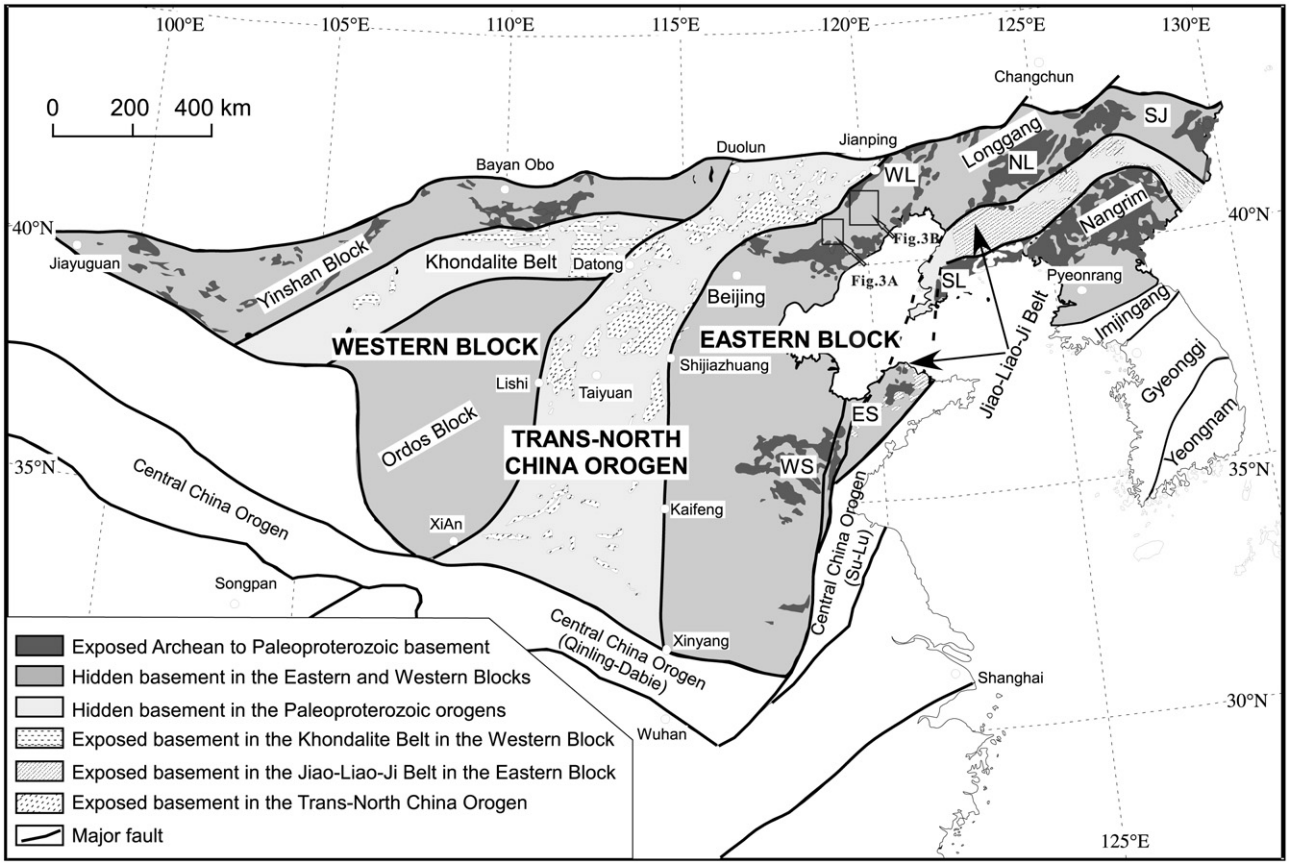


Fig. 1. Tectonic subdivision of the North China Craton (after Zhao et al., 2005).

ca. 1.35 Ga paleopoles are significantly different from each other, they suggest that the NCC may have remained in continuity with Baltica and Laurentia during the Mesoproterozoic. In particular, the quoted age of the Yangzhuang Formation is now problematic in the context of new results from the U–Pb dating method. A precise U–Pb zircon dating result (ca. 1.35 Ga) has recently been obtained by S.-H. Zhang et al. (2009) for sills emplaced into the Xiamaling and Wumishan formations overlying the Yangzhuang Formation (Fig. 2), suggesting that the ca. 1.35 Ga age assignment is too young for the underlying Yangzhuang Formation, and the significance of paleomagnetic poles obtained from this formation

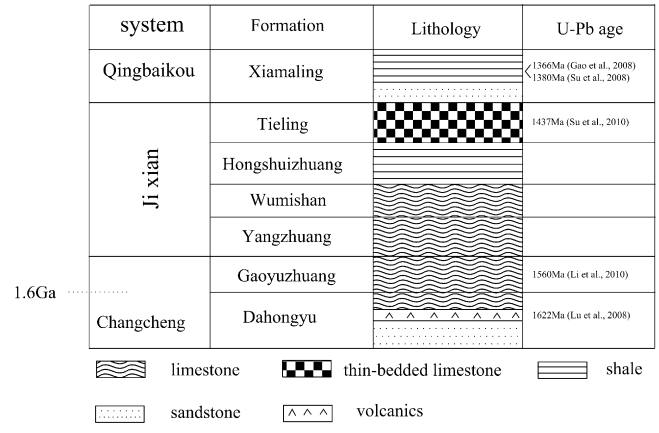


Fig. 2. Subdivision of Mesoproterozoic strata developed in the northern North China Craton with U–Pb age constraints (modified after S.-H. Zhang et al., 2009; Meng et al., 2011).

therefore need to be revised. In summary, the configuration of the NCC in the Columbia supercontinent is still actively debated and high quality Mesoproterozoic paleomagnetic data are clearly desirable. Here we report a ca. 1.35 Ga paleomagnetic pole from a suite of geochronologically dated Mesoproterozoic sills broadly intruding the Xiamaling and Wumishan formations in the northern NCC with an aim of providing a constraint to the configuration of the NCC in the Columbia supercontinent.

2. Geologic background

The NCC is divided into the Western and Eastern Blocks. The former is subdivided into the Yinshan Block in the north and the Ordos Block in the south, which may have amalgamated along the E–W trending Khondalite Belt at ~1.95 Ga (Fig. 1; Zhao et al., 2005; Xia et al., 2006a,b, 2008, 2009; Yin et al., 2009, 2011); whereas the latter is subdivided into the Longgang Block in the northwest and the Langrim Block in the southeast, separated by the Paleoproterozoic Jiao-Liao-Ji Belt (Fig. 1; Li et al., 2005, 2006, 2012; Lu et al., 2006, 2008; Li and Zhao, 2007; Zhou et al., 2008; Tam et al., 2011, 2012). The final amalgamation of the Eastern and Western Blocks to form the coherent basement of the NCC is considered to have occurred along the Trans-North China Orogen at about 1.85 Ga (Fig. 1; Zhao et al., 2000, 2001, 2010; Guo et al., 2002; J. Zhang et al., 2009; H. Li et al., 2010; S.Z. Li et al., 2010; Wang et al., 2010) and may have coincided with the global assembly of the Paleo-Mesoproterozoic Columbia supercontinent.

Following the final amalgamation of the NCC at ~1.8 Ga, a series of well-preserved unmetamorphosed or only weakly metamorphosed Precambrian sediments developed mainly over the central part of the NCC. These Paleo-Mesoproterozoic strata are

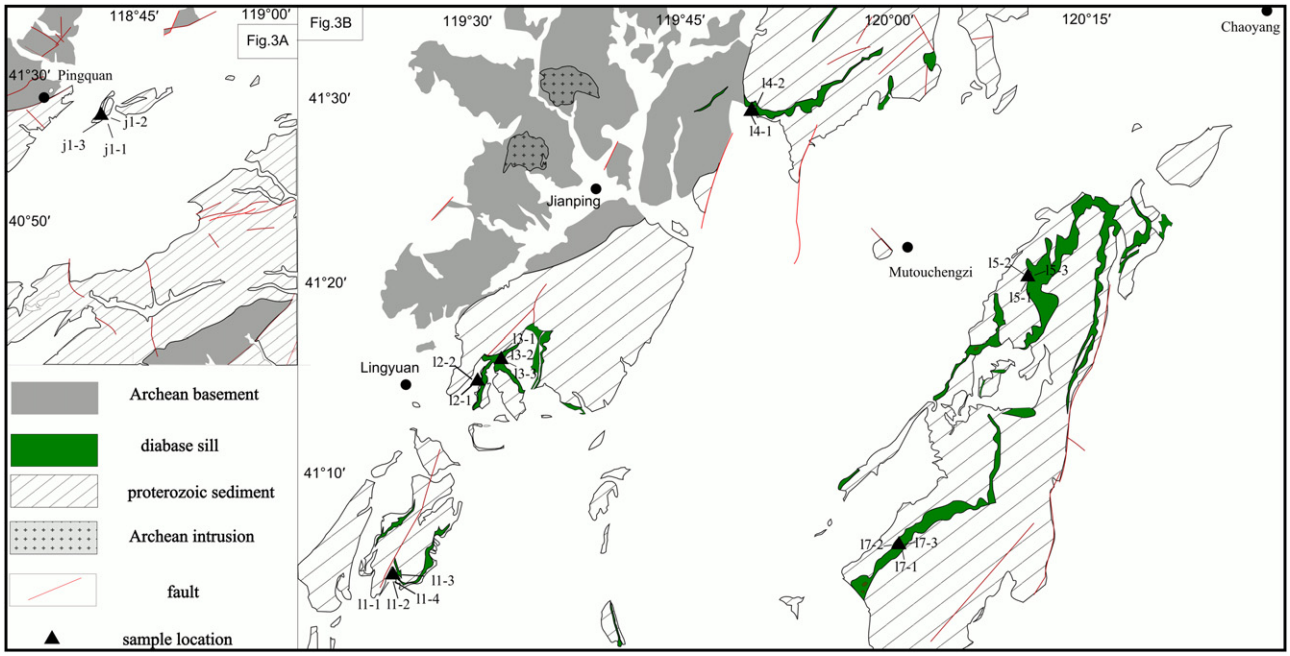


Fig. 3. Geological sketch map of the sampling areas; A and B represent geological sketches of sampling sub-areas in which black triangles indicate paleomagnetic sampling locations.

generally divided into four well-established systems, named the Changcheng, Jixian, Qingbaikou and Sinian systems which from the base to the top are composed mostly of clastic and carbonate rocks, considered to have developed in an extensional setting.

As summarized in Fig. 2, stratigraphic ages of these Paleo-Mesoproterozoic sediments have become more tightly controlled with the precise geochronological data obtained in recent years. For the Dahongyu Formation of the Changcheng System, a layer of volcanic rocks has yielded a U–Pb zircon age of ca. 1625 Ma though the origin of these volcanic rocks is debated (Lu et al., 2008). For the Gaoyuzhuang Formation of the Changcheng System, a tuff bed has been dated at ca. 1560 Ma using zircon U–Pb dating (H. Li et al., 2010), whilst the Tieling Formation at the top of the Jixian System was thought to have formed at ca. 1437 Ma based on geochronological dating on a K-bentonite bed (Su et al., 2010). The overlying Xiamaling Formation of the Qingbaikou System is now suggested to have formed around 1400 Ma ago, as demonstrated by zircon SHRIMP dating results of the tuff bed, K-bentonite in this formation (Gao et al., 2008; Su et al., 2008) and recent LA-ICP-MS dating of baddeleyite in diabase sills emplaced into it (Li et al., 2009; S.-H. Zhang et al., 2009). Indeed, these sills extend hundreds of kilometers from west to east and are widespread recorded in the northern area of the NCC, such as the Pingquan and Kuancheng areas of Hebei Province, and the Lingyuan, Chaoyang areas of Liaoning Province; some of them were even emplaced into the Wumishan Formation of the Jixian System. In particular, the zircon and baddeleyite dating technique on a diabase sample from the Chaoyang area of Liaoning province has yielded weighed mean $^{207}Pb/^{206}Pb$ ages of 1345 ± 12 Ma and 1353 ± 14 Ma (S.-H. Zhang et al., 2009). Furthermore, these sills intrude the host sedimentary rocks in a bed-parallel manner so that bedding attitudes of the sills can be well determined from the well-bedded host sedimentary rocks. The thickness of the sills varies from tens of meters to hundreds of meters and the lengths range from several kilometers to tens of kilometers.

In addition, our field observations reveal that these sills and the host sediments have been folded together and the folding time should be no earlier than the Mesozoic in which a large scale intraplate deformation, namely Yanshan movement, tremendously shaped the Eastern NCC, which is generally thought to result from continuous subduction of the Pacific plate into the juvenile Asian plate (Y. Zhao et al., 2004; Dong et al., 2007). Prior to the Mesozoic, the NCC remained stable and experienced little magma activities from the Mesoproterozoic to the early time of the Mesozoic (Zhai et al., 2004).

3. Paleomagnetic methods

The paleomagnetic samples were all collected from geochronologically precisely dated sills intruded into the Xiamaling and Wumishan formations. Locations of the drilling sites are generally distributed in very fresh rock outcrops in streambed or in artificial quarries, with some sites in natural outcrops. The core samples were collected using a portable gasoline-powered drill, and oriented by magnetic compass, or both magnetic and sun compasses in sunny days. The distribution of paleomagnetic sampling sites is shown in Fig. 3.

In the laboratory samples were first cut into standard cylindrical specimens. In order to identify the main magnetic minerals we determined magnetic susceptibility versus temperature measurements on pilot samples using a KLY-3S instrument in which sample powder was heated in air from room temperature up to 700 °C and cooled back to room temperature. For the purpose of determining the size of magnetic minerals in the samples, hysteresis loops were observed in the pilot samples by a Micromag 3900 Vibrating Sample Magnetometer with maximum magnetic field of 1.0 T.

Scanning electron microscope (SEM) and light microscope analysis were also employed to further identify the nature of the magnetic minerals. Finally, in terms of the rock magnetic results for pilot samples, all the specimens were subject to progressive thermal demagnetization using temperature intervals up to the maximum unblocking temperatures of the remanence carriers. The remanence was measured using a 2G-755 cryogenic magnetometer in a magnetically shielded house. Demagnetization results were processed using principle component analysis

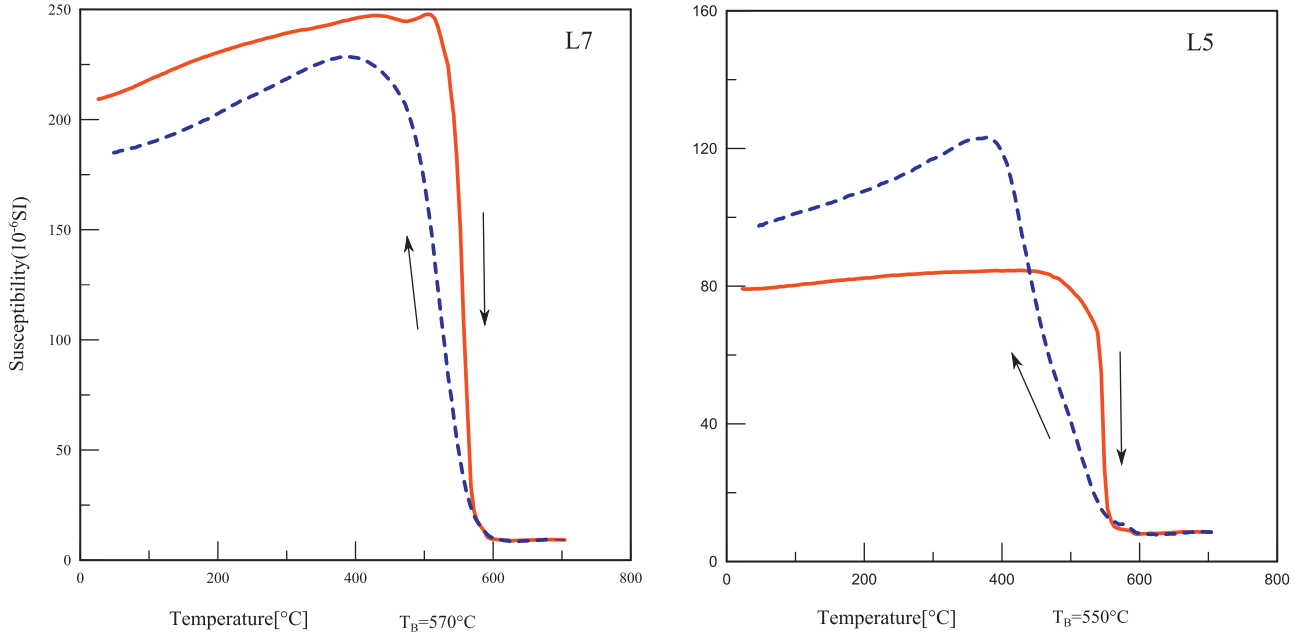


Fig. 4. Thermomagnetic curves (susceptibility versus temperature) for pilot samples of the sills. Solid and dash lines indicate heating and cooling curves, respectively. T_B denotes the unblocking temperature at which the susceptibility declined to the minimum.

(Kirschvink, 1980) and interval-mean results were evaluated by Fisher statistics (Fisher, 1953). All the magnetic measurements and processing were preformed in the Paleomagnetism and Geochronology Laboratory (PGL) of the Institute of Geology and Geophysics, Chinese Academy of Science (IGGCAS).

4. Results

4.1. Rock magnetic results

Pilot samples from the sills were ground into the powder in a mortar and then subject to thermomagnetic and magnetic hysteretic property analysis. Representative results are presented in Fig. 4. We can see clearly that magnetic susceptibility of sample L7 drops dramatically at nearly 570 °C during the heating process, whilst the other sample (sample L5) has a slightly lower unblocking temperature around 550 °C. Such a phenomenon implies that Ti-poor titanomagnetite is the main magnetic mineral in these sills.

To illustrate hysteresis property of the sill samples, two out of eleven measured samples are selected (Fig. 5A, B). It can be seen that they are all saturate by 0.4 T with coercivities of around tens of milliTesla (mT). Since (titano)magnetite saturates between 0.3 and 0.5 T (David, 1983), these behaviors also support the interpretation that (titano)magnetite is the remanence carrier in these sills. According to the Day diagram (Day et al., 1977; Dunlop, 2002) which can be used to recognize the domain type, the hysteresis parameters for the pilot samples show that the magnetic particles are all located in the pseudo-single domain (PSD) range (Fig. 5C).

Meanwhile, the SEM and light microscope analyses also show straightforward evidence for the presence of primary Ti-magnetite in the samples. As shown in Fig. 6A–C, Ti-magnetites can be very easily identified because of their sizes reaching several hundred micrometers; such large sized mineral grains are usually formed by slow cooling at depth in the crust. In thin- and polished rock sections viewed under light microscopy, we observe that the Ti-magnetites, pyroxenes and plagioclases have clear ophitic intergrowth relationship (Fig. 6D, E) indicative of an original magmatic texture without chemical alteration by fluids or later weathering. This implies that the magnetic minerals were crystallized from the original magma and thus likely to preserve primary and stable remanence acquired during cooling accompanying initial emplacement.

4.2. Paleomagnetic results

All the specimens were subject to thermal demagnetization up to 560 °C using a TD-48 demagnetizer employing initial temperature intervals of 50 °C from room temperature to 500 °C, and subsequent steps as narrow as 10–20 °C up to the Curie points. From demagnetization results exhibited on the orthogonal diagrams (Zijderveld, 1967), we can firstly resolve a low temperature component unblocked mostly before ~300 °C (Fig. 7). Collectively, this low temperature component approximately parallels the present day geomagnetic field direction at the sampling locality and is therefore interpreted as an overprinted viscous remanence. Following removal of this low temperature component, a high temperature component decaying almost linearly to the origin is isolated from the large majority of demagnetized sill specimens. As shown in Fig. 7, the characteristic high temperature remanence has almost been unblocked by temperatures up to ~560 °C. This feature is compatible with the rock magnetic results from pilot sill samples showing that Ti-poor magnetite is the main remanence carrier in these sills. The high temperature characteristic remanence seems to have a single reverse polarity with a northwesterly declination and shallow upward inclination following tilt adjustment (Fig. 7a–l) with only one specimen from site L1-3 showing approximately antipodal direction (Fig. 7m–o). Among the 20 sampling sites, sites L2-1 from the east of the city of Lingyuan (Fig. 3) shows an abnormal characteristic direction both before and after tilt adjustment (Fig. 8A and Table 1), whilst site L3-1 shows no stable demagnetization behavior and no site-mean direction isolated here. Excluding these two sites, the remaining 18 sites yield a group-mean direction of $D = 292.7°$, $I = -9.4°$ with $\alpha_{95} = 7.8°$ before, and $D = 294.4°$, $I = -31.7°$ with $\alpha_{95} = -4.3°$ after tilt correction (Table 1). As shown in Fig. 8A, the data grouping is significantly improved after tilt correction with Fisherian precision parameter increasing from ~21 to 64. Further using the Watson and Enkin (1993) fold test presents an optimal concentration of the characteristic site-mean directions at

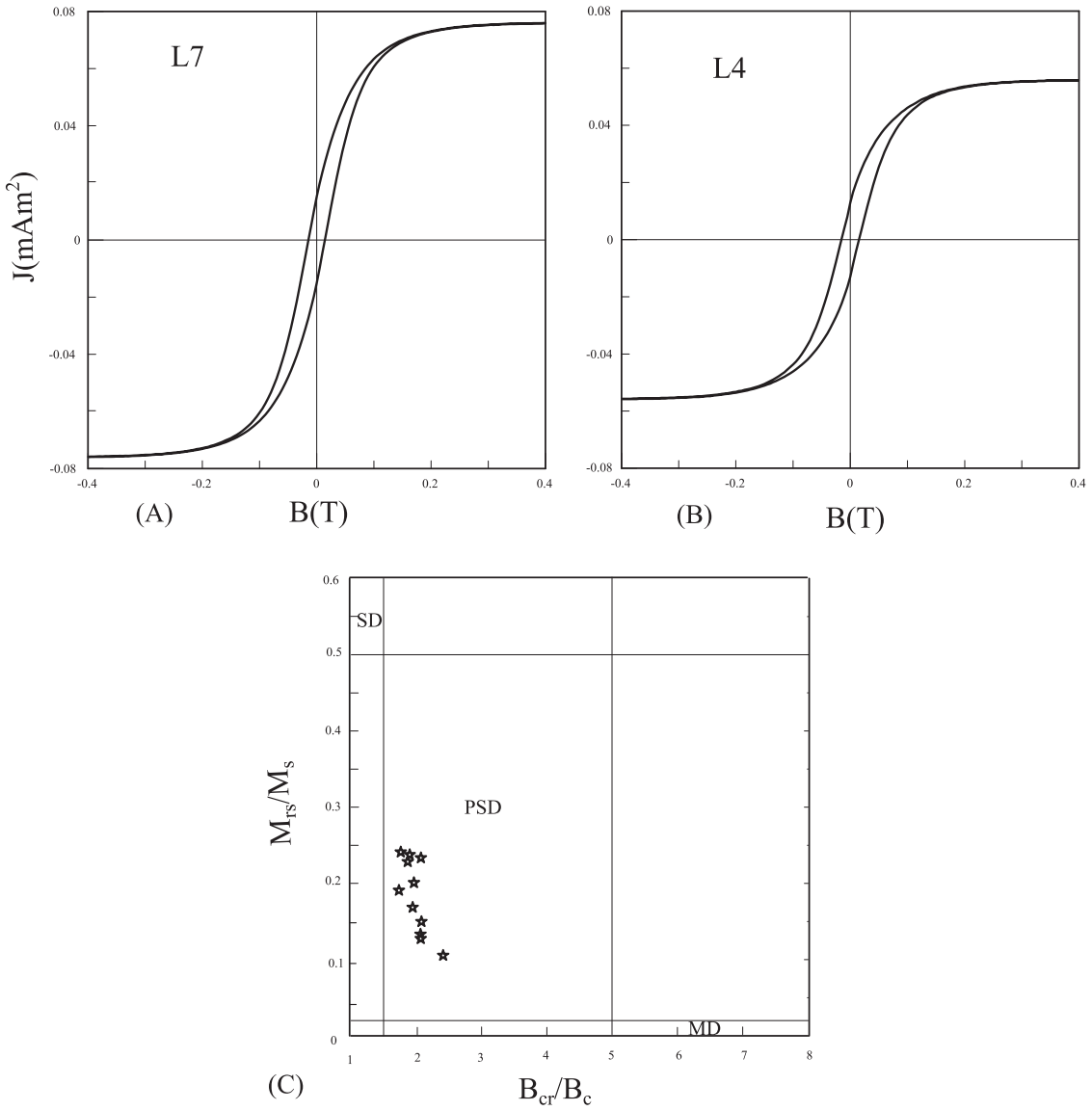


Fig. 5. Magnetic hysteresis loops for two representative sill samples: (A) sample L7 and (B) sample L4; (C) Day diagram, in which Mr and Ms are saturation remanence and saturation magnetization; Bc and Bcr are coercivity and coercivity of remanence; SD: single domain, PSD: pseudo single domain, MD: multidomain.

86.2% unfolding with a 95% confidence limit of 4.2% (Fig. 8B). Application of McFadden (1990) fold test identifies a positive result with definition 2 of SCOS of 11.189 before and of 3.997 after tilt correction; the statistical threshold of SCOS is 4.940 at 95% confidence level.

5. Discussion

Noting that both rock magnetic experiments and thermal demagnetization behaviors indicate that the characteristic remanence carried by Ti-poor titanomagnetites in PSD state, whilst the SEM and light microscopic observations show that the sill samples have an original magmatic texture indicating that Ti-poor titanomagneties are likely to be the original ferromagnets in the absence of chemical alternation and strong tectono-thermal events, we are able to interpret the characteristic remanence as a primary magnetization. Furthermore the positive fold test results (McFadden, 1990; Watson and Enkin, 1993) indicate a pre-folding origin with the characteristic remanence acquired before the time of regional folding. The corresponding paleomagnetic pole at $\lambda = 5.9^{\circ}$N, $\varphi = 359.6^{\circ}$E with $A_{95} = 4.3^{\circ}$ is notably different from Early Paleozoic and post-Permian paleopoles from the NCC (Huang et al., 1999, 2008; Yang et al., 2002), further supporting the view that the isolated characteristic remanence is primary. Since the magnetizations are almost of one polarity it is important to assess whether the computed paleopole has sufficiently averaged geomagnetic field secular variation. To address this point, we calculate an angular standard deviation (ASD) of the observed virtual geomagnetic poles (VGPs) and determine lower and upper 95% confidence limits of the ASD using Vandamme's (1994) method, which yields an ASD of 10.0° with $S_l = 8.1^{\circ}$ and $S_u = 13.0^{\circ}$. Although paleosecular variation in the Proterozoic is not well understood it is therefore likely that the paleopole obtained from these ca. 1.35 Ga sills, which are broadly distributed within the northern areas of the NCC, will have sufficiently smoothed out the paleosecular variation because a compatible ASD value for paleolatitudes around 10–20° is found in the Cretaceous normal superchron (Biggin et al., 2008) and a low reversal frequency of the geomagnetic field seems to have prevailed during Precambrian times (Dunlop and Yu, 2004). More recently, Linder and Gilder (2012) demonstrate that the latitude dependency of the ASD may stem from a mathematical artifact so that the directional populations better qualify the

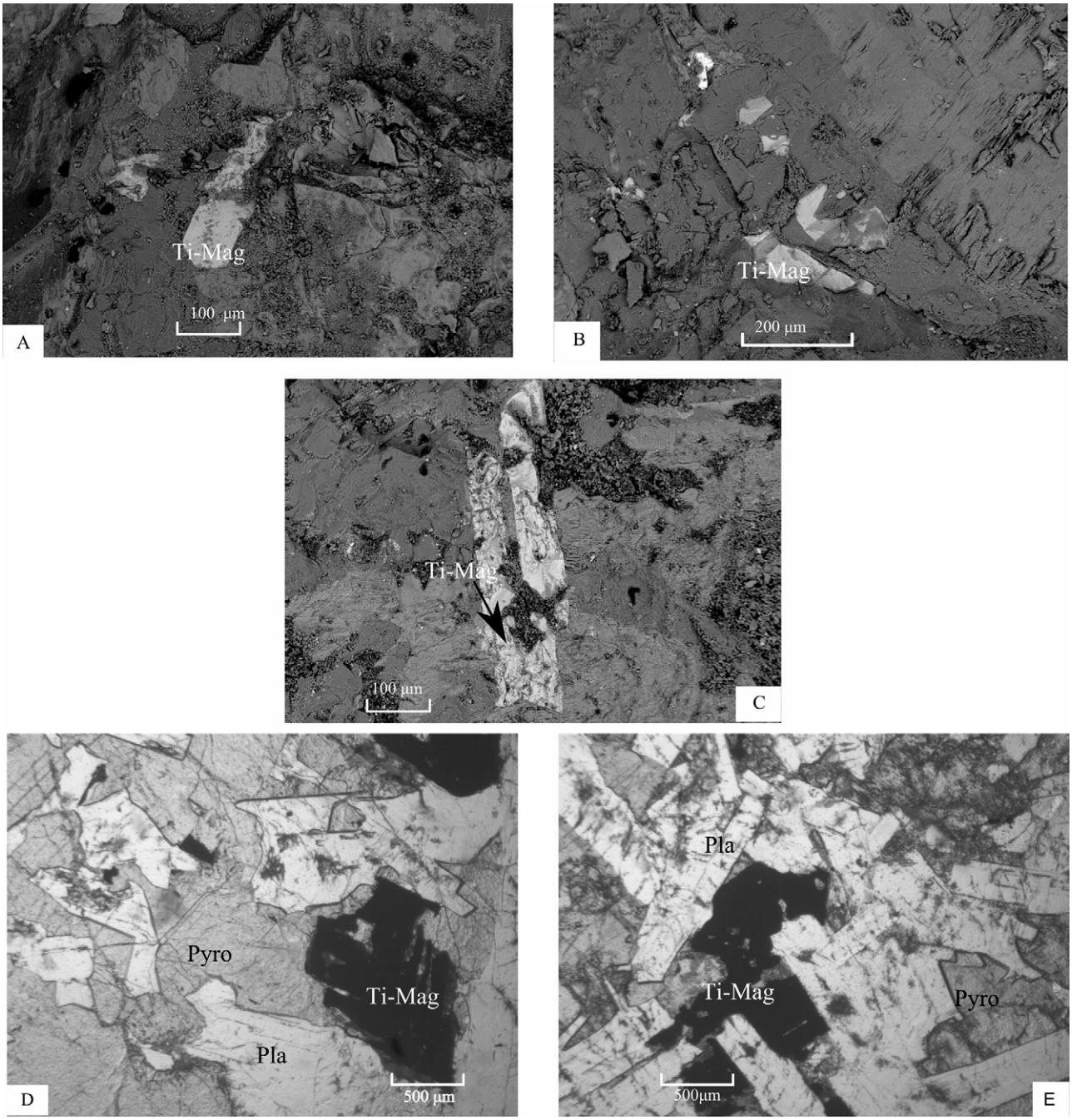


Fig. 6. The SEM images (A–C) and light microscopy (D and E) results for pilot sill samples. Pla, plagioclase; Pyr, pyroxene; Ti-mag, Titanomagnetite.

field secular variation. However, directional ASD calculated here, of 10.1° with S_l = 8.2° and S_u = 13.2°, does not show significant difference with that for the VGPs. The sills sampled for this study are all precisely U–Pb dated to be ca. 1.35 Ga (Gao et al., 2008; Li et al., 2009; S.-H. Zhang et al., 2009) reinforcing the view that the paleomagnetic pole obtained here is representative for the NCC at ca. 1.35 Ga and can be used for paleogeographic reconstructions during the Mesoproterozoic NCC.

So far, there are only few available paleomagnetic results reported for the Paleo-Mesoproterozoic NCC where Halls et al. (2000) firstly reported a late Paleoproterozoic paleopole from the ca. 1.78–1.76 Ga dyke swarm developed in the Taihang Mountains of the Trans-North China Orogen and proposed that the North China, Laurentia and Baltica cratons may have been connected with each other by ca. 1.8 Ga. Most recently, Piper et al. (2011) demonstrated that new paleomagnetic data of these dyke swarms, which developed over much wider areas of the Trans-North China Orogen and support the Zhao et al. (2003b) configuration in which the NCC was linked to the Indian Shield. Zhang et al. (2012) obtained also a ~1.78 Ga pole from the Xiong'er volcanics which widely outcropped in the southern NCC. On the other hand, Pei et al. (2006) and Wu et al. (2005) reported two high quality paleopoles from the Yangzhuang sediments in the lower of the Jixian System, which was previously thought to have formed around ca. 1.35 Ga (Wang et al., 1995). Both of these two paleopoles suggest that the NCC was the member of the Columbia supercontinent during the Yangzhuang Formation although they are separated by an angular distance of 27.4 ± 9.8° (Table 2 and Fig. 9). However, as pointed out before,

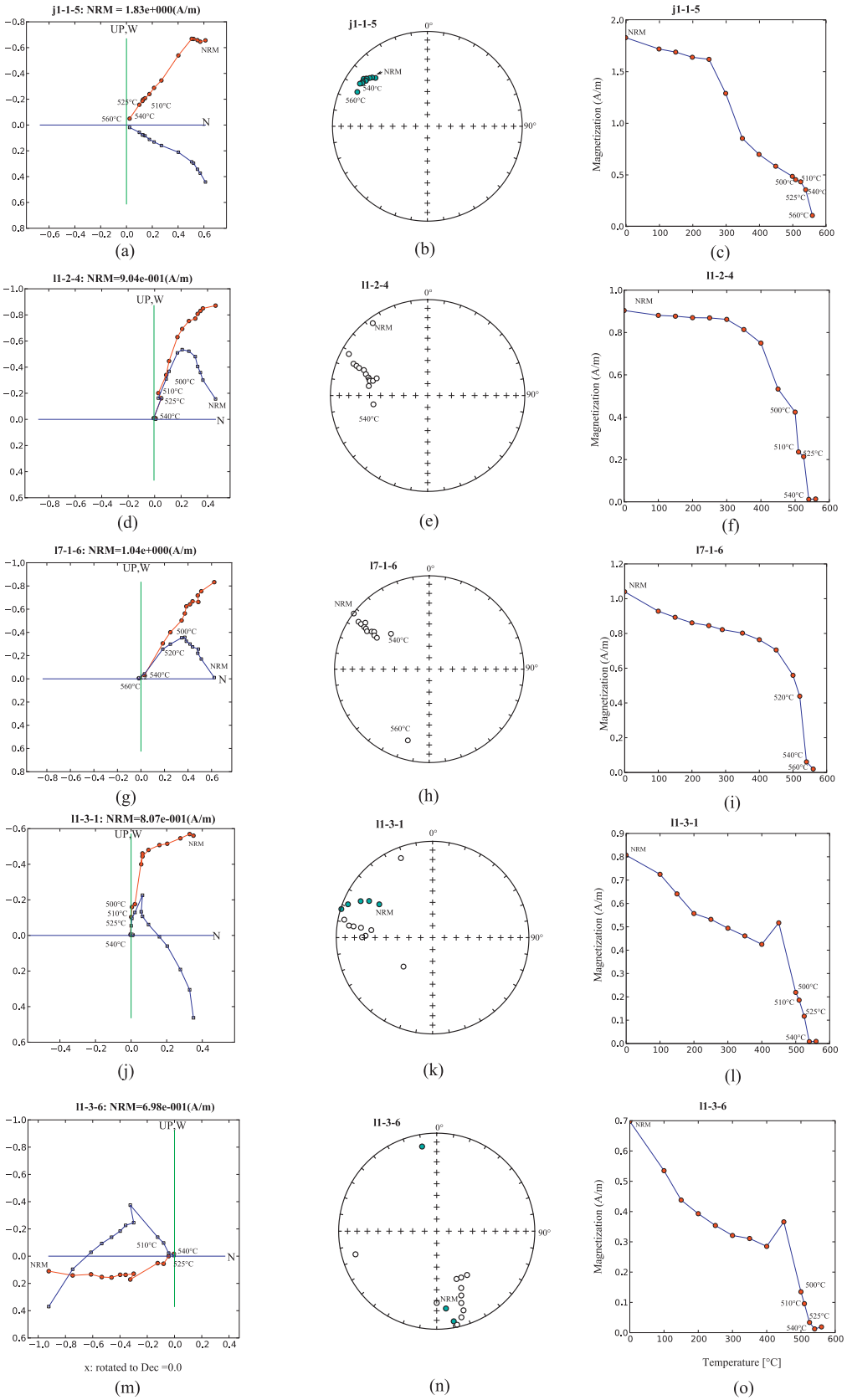


Fig. 7. Thermal demagnetization treatment for five representative sill specimens. Left plots: directional orthogonal demagnetization diagram; the red and blue lines represent the declination and inclination of remanence, respectively; middle column: equal-area projections of remanence variations with demagnetization temperatures; right column: intensity decay diagrams. All the orthogonal plots and equal-area projections are drawn in geographic coordinates. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

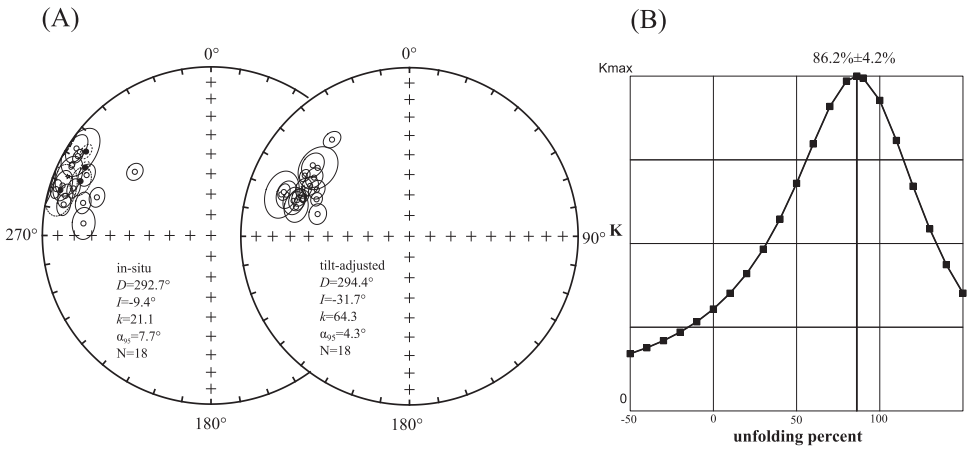


Fig. 8. (A) Equal-area projection of the site-mean observations before tilt correction; the star in the plot denotes the Fisherian mean direction. (B) Increment unfolding analysis using Watson and Enkin (1993) method.

the sedimentary age of this formation might be as early as ca. 1.5 Ga (S.-H. Zhang et al., 2009) in the light of recent U–Pb dating results on the sills intruding both the overlying Wumishan and the Xiamaling formations of the Qingbaikou System (H. Li et al., 2010; Su et al., 2010). In addition, Zhang et al. (2006) reported also two preliminary paleopoles comprising a pole from the Yunmengshan Formation dated ca. 1260 Ma by the Rb–Sr method and the ca. 1200 Ma (estimated) pole from the Baicaoping Formation from the southern NCC. These two paleopoles both have positive reversal and fold test results so that they seem to have high quality and can be used to further constrain the geographical relationship between the NCC and surrounding shields. However, recent LA-MC-ICPMS zircon U–Pb dating of an interbedded tuffite layer in the middle part of the Luoyukou Formation, the uppermost of the Luoyu Group, presents a precise age of 1611 ± 8 Ma, suggesting that the top of the Luoyu Group should have a depositional age around ~1600 Ma (Su et al., 2012). In this case, the underlying Baicaoping and Yunmengshan formations, the lower two members of the Ruyang Group, developed in the southwestern Henan Province should have a depositional age older than ~1600 Ma rather than around ~1200–1300 Ma. Because of this dispute, we exclude these two paleomagnetic poles from further discussion.

Owing to the lack of successive polarity sequence for the Precambrian geomagnetic field, we tentatively regard the ca. 1.35 Ga paleopole at $\lambda = 5.9°$S, $\varphi = 179.6°$E with $A_{95} = 4.3°$ as of normal polarity in the light of the principle of least apparent polar wander (APW) path from this ca. 1.35 Ga paleopole to the Yangzhuang Formation pole (Pei et al., 2006). Hence the NCC is accordingly suggested to have been located at ~18°N in the Northern Hemisphere during the emplacement of the sills and this further supports the view that the NCC remained in low latitudes during the possible later history of the Columbia supercontinent (Halls et al., 2000; Pei et al., 2006; Wu et al., 2005). Thus, we can combine currently available ca. 1.76–1.78 Ga, 1.5 Ga and 1.35 Ga paleopoles to construct a Paleo-Mesoproterozoic APW path for the NCC, which

Table 1
Summary of sampling information and results of characteristic remanence from the ca. 1.35 Ga sills in the North China Craton.

Site ID	λ_s (°N)	φ_s (°E)	Strike/Dip	n/N	D_g (°)	I_g (°)	D_s (°)	I_s (°)	k	α_{95} (°)	φ_p (°E)	λ_p (°N)
j1-1	40.9688	118.7462	248/65	9/10	303.9	11.7	291.8	−40.5	298.5	3.0	357.4	0.0
j1-2	40.9691	118.7458	248/65	5/10	298.4	16.2	290.7	−33.6	908.5	2.5	1.5	2.7
j1-3	40.9692	118.7458	248/65	8/9	292.6	17.2	287.0	−28.9	92.7	5.8	6.0	2.2
l1-1	41.0680	119.4172	151/39	5/10	282.4	−12.6	297.3	−39.1	338.3	4.2	355.1	4.2
l1-2	41.0680	119.4181	178/20	9/11	285.6	−12.4	288.3	−31.4	129.8	4.5	4.7	2.0
l1-3	41.0701	119.4184	120/17	5/10	275.8	−25.9	285.0	−31.9	140.2	6.5	6.7	−0.5
l1-4	41.0700	119.4176	180/30	9/10	287.3	6.5	288.6	−22.1	29.6	9.6	8.5	6.0
l2-1*	41.2643	119.5322	97/21	10/12	317.3	−16.6	320.0	−2.5	43.2	7.4	350.5	34.1
l2-2	41.2646	119.5322	97/21	8/9	310.7	−41.2	322.0	−27.7	189.4	4.0	340.2	23.9
l3-1	41.2664	119.5846	163/26	0/7	No available data							
l3-2	41.2657	119.5845	191/26	8/10	303.5	−6.4	307.1	−30.2	99.7	5.6	351.9	14.5
l3-3	41.2654	119.5874	191/26	6/10	303.2	−9.4	307.5	−33.2	48.0	9.8	350.4	13.3
l4-1	41.4904	119.8563	133/14	10/11	289.4	−30.5	298.0	−35.2	134.8	4.2	356.8	6.4
l4-2	41.4837	119.8592	133/32	9/10	285.2	−24.0	302.7	−34.8	136.1	4.4	353.6	9.5
l5-1	41.3323	120.1856	196/16	9/10	290.0	−16.6	290.6	−32.5	180.4	3.8	3.4	2.9
l5-2	41.3328	120.1850	196/16	8/10	289.6	−9.8	289.9	−25.7	95.1	5.7	6.9	5.4
l5-3	41.3324	120.1825	169/20	8/10	287.8	−6.5	290.6	−23.8	270.1	3.4	7.2	6.6
l7-1	41.0976	120.0191	216/20	11/11	301.7	−10.4	301.1	−30.3	290.0	2.7	356.8	10.9
l7-2	41.0995	120.0227	239/30	8/10	297.7	−10.2	290.4	−35.0	1169.7	1.6	2.3	1.8
l7-3	41.0986	120.0251	239/30	7/11	296.9	−20.6	284.7	−44.5	176.6	4.6	1.0	−6.8
			Mean	(18/20)	292.7	−9.4			20.7	7.8	359.6	5.9
							294.4	−31.7	65.5	4.3	$K = 66.5$,	$A_{95} = 4.3$

Abbreviations: n/N: number of specimens used for calculation/number of specimens demagnetized; Strike/Dip: azimuth of right-hand strike and angle of dip of the sills; D_g and I_g (D_s and I_s) indicate declination and inclination before (after) tilt correction. α_{95} is Fisherian statistic 95% confidence; k is the precision parameter; λ_s and φ_s (λ_p and φ_p) represent latitude and longitude of the sampling site location (calculated virtual geomagnetic poles in stratigraphic coordinates); the site marked by '*' is the one with an abnormal remanence.

The Fisherian mean direction is calculated after converting all the site-mean directions to the sampling location of site j1-1.

* Site has been excluded from final Fisherian averaging because of its large deviation from the majority of the collection.

Table 2
Available Paleo-Mesoproterozoic paleopoles from Laurentia, Baltica, Siberia, India and the North China Craton.

Age (Ma)	Pole	Formation	P_{lat} (°N)	P_{long} (°E)	A_{95} (°)	Test	References
	Baltica						
1780	Shok	Vazhinka River Fm	39.0	217.0	7.0	R	1
1452	Lad	Ladoga intrusions	15.2	177.1	6.0	C	2
1384	MS	Mashak Fm	1.8	193.0	13.0	R	3
1265	PJot	Post-Jotnian mean	4.0	158.0	4.0	C	4
	Laurentia						
1740	Cleav	Cleaver dikes	19.4	263.0	6.1	C	5
1476	StFr	St. Francois Mtns	−13.0	219.0	6.0	G, F, C	6
1382	Zig	Zig–Zag Dal and intrusions	11.0	229.0	3.0	R	7, 8, 9
1267	Mac	Mackenzie mean	4.0	190.0	5.0	C	10
	North China						
1780	XR	Xiong'er volcanics	50.2	263.0	4.5	R	11
1769	TH	Taihang dykes	36.0	247.0	4.0	C	12
1500?	YZ(a)	Yangzhuang Fm	17.0	214.5	6.0	R	13
1500?	YZ(b)	Yangzhuang Fm	2.4	190.4	12.0	R	14
1350	LH	Liaoning, Hebei sills	−5.9	179.6	4.3	F	This study
	Siberia						
1473	Olen	Olenëk mafic intrusions	−33.6	73.1	10.4	C	15
1384	Chier	Chieress dike	−4.0	78.0	7.0	–	16
	India						
1880	Bastar	Dykes from Bastar craton	31	330	11.9	R	17

Notes: P_{lat}, P_{long}: latitude and longitude of paleomagnetic poles; A_{95}: the 95% confidence radius of paleopoles. References: 1 = Pisarevsky and Sokolov (2001), 2 = Lubnina et al. (2010), 3 = Lubinina (2009), 4 = Buchan et al. (2000), 5 = Irving et al. (2004), 6 = Meert and Stuckey (2002), 7 = Marcussen and Abrahamsen (1983), 8 = Abrahamsen and Van der Voo (1987), 9 = Upton et al. (2005), 10 = Buchan and Halls (1990), 11 = Zhang et al. (2012), 12 = Halls et al. (2000), 13 = Wu et al. (2005), 14 = Pei et al. (2006), 15 = Wingate et al. (2009), 16 = Ernst et al. (2000), 17 = Meert et al. (2011).
C: contact test; R: reversal test; F: fold test; G: conglomerate test.

can provide constraints on the paleogeographic evolution and kinematic features of the NCC during the breakup of Columbia. For this purpose, we have compiled all reliable Paleo-Mesoproterozoic paleomagnetic poles from the cratonic Laurentia, Baltica, Siberia and India (Table 2).

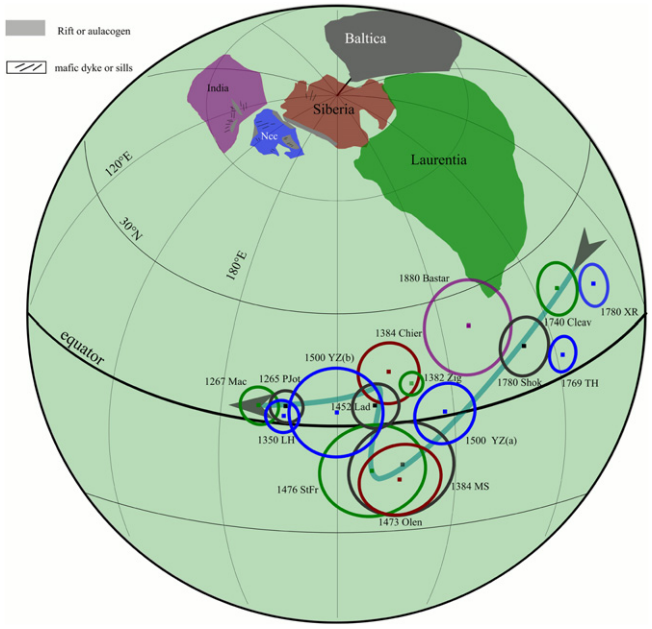


Fig. 9. Available Paleo-Mesoproterozoic paleomagnetic poles from Laurentia, Baltica, Siberia, North China, and India and the suggested paleogeographic reconstruction showing connection of the North China Craton (NCC) with the Laurentia–Baltica–Siberia (LBS) assembly and the Indian Craton in the Columbia supercontinent, in which cratonic Laurentia is fixed to present coordinates. All paleopoles are rotated to present Laurentia reference frame. The Euler rotation parameters are: Baltica, 47.5°N, 1.5°E, +49° (Evans and Pisarevsky, 2008); Siberia, 78.0°N, 99.0°E, +147° (Evans and Mitchell, 2011); NCC, 25.0°N, 21.0°E, +40.0°; India, 49.3°N, 64.6°E, +117.0°. The green, gray, blue, red, and purple color symbols indicate paleopoles from Laurentia, Baltica, NCC, Siberia, and India, respectively.

For Laurentia, due to the lack of a ca. 1.77 Ga paleomagnetic pole, the 1.74 Ga paleopole from the Cleaver Lake Dykes can be used based on the baddeleyite U–Pb dating (Irving et al., 2004), whilst for the ~1.5 Ga, the ca. 1476 Ma St. Francois Mountains igneous province result is an appropriate choice (Meert and Stuckey, 2002). For ~1.35 Ga, the paleopole resolved from the Zig Zag Dal basalts and Victoria Fjord dikes dated ca. 1382 Ma (Abrahamsen and Van der Voo, 1987; Marcussen and Abrahamsen, 1983; Upton et al., 2005) has high quality and can be used for construction during the Mesoproterozoic APW path for the craton. In addition, the 1267 ± 2 Ma paleopole from the Mackenzie dykes has high quality and can be used to address paleogeographic relationship between Laurentia and Baltica (Buchan and Halls, 1990). For Baltica, we have selected four similar aged poles which are 1780 Ma key pole (Pisarevsky and Sokolov, 2001) from the Vazhinka River succession, the 1452 Ma pole from the Ladoga intrusions (Lubnina et al., 2010), the 1384 Ma pole from the Mashak Formation (Lubnina, 2009), and the 1265 Ma pole from the Postjotnian intrusions in western Finland (Buchan et al., 2000). However, for Siberia, there are no available paleomagnetic poles reported for ca. 1.8 Ga time and only two Mesoproterozoic paleopoles can be incorporated into our further discussion; these are the ca. 1473 Ma paleopole from the Olenëk mafic intrusions (Wingate et al., 2009) and the 1384 Ma pole from the Chieress dikes (Ernst et al., 2000). These two Mesoproterozoic paleopoles have been repeatedly used to constrain paleogeographic relationship between Siberia and surrounding cratons (Evans and Mitchell, 2011; Wingate et al., 2009). For India, the high quality Paleo-Mesoproterozoic poles are very sparse and only a preliminary Paleoproterozoic paleopole was reported from the ~1.88 Ga dykes in the Bastar and Dharwar cratons (Meert et al., 2011).

In the light of these selected Paleo-Mesoproterozoic paleopoles for Laurentia, Baltica, Siberia, and India, it is clear that Laurentia and Baltica may have had a tight connection during evolution of the Columbia supercontinent. As proposed by Buchan et al. (2000), Lubnina et al. (2010), and Salminen and Pesonen (2007), Laurentia and Baltica have compatible APW paths between ca. 1.74 and 1.27 Ga (Fig. 9), suggesting that the Laurentia–Baltica connection

may not been broken down until at least ca. 1.3 Ga. For Siberia, in spite of insufficiency of available Paleo-Mesoproterozoic paleopoles, paleomagnetism cannot provide direct constraints on its paleogeographic scenario at ca. 1.8 Ga. However, the Siberian Craton might have been adjacent to Laurentia before ca. 1.27 Ga (Evans and Mitchell, 2011) in terms of the concordance of ca. 1.5–1.4 Ga paleopoles between Laurentia and Siberia (Fig. 9). Therefore, it is very likely that the Laurentian, Baltic, and Siberian shields were united as a coherent block between ~1.8 and 1.3 Ga to constitute one of the major nuclei of the Columbian supercontinent. Nonetheless, as mentioned above, the paleogeographic position of the NCC in the Columbia supercontinent is disputable because of both geological complexity and inaccuracy of Paleo-Mesoproterozoic APW path so that several candidates such as the cratonic Baltica, Siberia, as well as India have been proposed to be linked to the NCC. As shown in Fig. 9, the newly constructed Paleo-Mesoproterozoic APW path for the NCC has, on the whole, a quite similar trend compared with that for Laurentia between 1.78 and 1.35 Ga. In particular, the 1.35 Ga and two Yangzhuang poles (Pei et al., 2006; Wu et al., 2005) fall close to their predicated locations between 1350 and 1500 Ma according to Piper's analysis results (Piper, 2010). These suggest that the NCC was very likely a member of the Laurentia–Baltica–Siberia (LBS) block by ca. 1.35 Ga. In addition, we note that Wu et al. (2005) and Pei et al. (2006) obtained distinctively different paleopoles from the ca. 1.5 Ga Yangzhuang Formation. At present we cannot identify which of these paleopoles is the most reliable, though we suspect that the paleopole reported by Pei et al. (2006) may represent a much younger record of geomagnetic field because it is essentially located near the ~1.4–1.3 Ga paleopoles of the LBS (Fig. 9).

In our paleogeographic reconstruction, the northern margin of the NCC is suggested to have been situated near to the western Siberia in the period 1.8–1.3 Ga in the light of the well-matched Paleo-Mesoproterozoic APW paths between the LBS and the NCC. This is consistent with several lines of geological evidence. Firstly, 1.4–1.3 Ga mafic intrusions have been discovered extensively in North America, Greenland, Siberia, Baltica, Antarctica, Australia and North China (Ernst et al., 2008). Especially in Siberia, a dyke swarm from the Anabar Shield has been dated 1384 ± 2 Ma by baddeleyite U–Pb method (Ernst et al., 2000). On the other hand, ~1.3–1.2 Ga carbonates, sills and mafic dykes have also been discovered from the Zhaertai–Bayan–Obo–Huade rift belt along the northern margin of the NCC (Yang et al., 2011). These approximately simultaneous magmatic events suggest that these cratons might have drifted together before ca. 1.4–1.3 Ga although we stress that they could each have originated from quite different large igneous province (LIP) events. Secondly, Li et al. (1996) correlated the Changcheng and Jixian systems in the NCC with the Riphean sediments in the western margin of the Siberian Craton, which may represent a Meso-Neoproterozoic passive margin (Pisarevsky and Natapov, 2003). This indicates that these two cratons might have been joined at that time.

For paleogeographic position of the Indian Craton in the Columbia supercontinent, we tentatively use the ca. 1.88 Ga paleomagnetic pole from the Indian Shield (Meert et al., 2011) to put the Indian Craton in the proximity of the NCC (Fig. 9). Noting that the Indian Craton is still suggested to connect with the NCC; however, this preliminary paleomagnetic data-constrained model is a little different with those proposed by previous studies (e.g. G. Zhao et al., 2004; Hou et al., 2008b; Piper et al., 2011; Zhang et al., 2012), in which the present day eastern margin of India is suggested to connect with the southern margin of the NCC. This scenario is mostly based upon approximately simultaneous magmatic events developed in the eastern India and the southern North China as well as a mantle plume hypothesis for the Xiong'er volcanic rocks developed in the southern margin of the NCC (Peng et al., 2005; Peng, 2010). As pointed out by Hou et al. (2008b), the ca. 1.8–1.9 Ga dyke swarms are widely developed in the Indian and North China cratons. In particular, large ca. 1.8 Ga dyke swarms were widely emplaced into the Trans-North China Orogen and adjacent areas in the central part of the NCC. These dykes as well as the Xiong'er volcanic rocks (dated ca. 1.76–1.78 Ga) along the southern margin of the NCC are considered to have resulted from a LIP formed by a mantle plume, though He et al. (2008, 2009, 2010a,b) and Zhao et al. (2009, 2011) argued that the Xiong'er volcanic rocks developed at an Andean-type continental margin arc. Peng and his colleagues (Peng et al., 2005; Peng, 2010) further argued that the dyke swarms developed in the central part of the NCC are the feeders of magmas forming the Xiong'er volcanic rocks and there is a missing part may have existed outside the southern NCC.

6. Conclusions

Positive fold test result and original magmatic texture of the collected sill samples indicate a primary nature of the characteristic remanence yielding a corresponding paleomagnetic pole at $\lambda = 5.9^{\circ}$S, $\varphi = 179.6^{\circ}$E with $A_{95} = 4.3^{\circ}$ ($N = 18$) for the NCC during ca. 1.35 Ga time. This ca. 1.35 Ga paleopole indicates a low latitude situation of around 18°N for the NCC. A newly constructed ca. 1.78–1.35 Ga APW path for the NCC matches very well with that for the Laurentia–Baltica–Siberia (LBS) when the NCC is rotated to a site near Siberia. This indicates that the NCC may have a similar kinematic evolution as the LBS and drifted together with Siberia during the period from ca. 1.8 Ga to ca. 1.35 Ga. The Indian Craton may also locate near the southern NCC in the light of their approximately simultaneous magmatic activities and a mantle plume hypothesis. On the other hand, the ca. 1.3–1.2 Ga magmatic events which occurred simultaneously in the North China and Siberian cratons may have resulted in complete separation between the NCC and Siberia. Nevertheless, more comprehensive investigations are desirable for improving the number and quality of paleopoles for the period from ca. 1.8 Ga to ca. 1.35 Ga so that the relationship of the NCC to surrounding cratons during evolution of the Columbia supercontinent can be better understood.

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References

Abrahamsen, N., Van der Voo, R., 1987. Palaeomagnetism of middle Proterozoic (c. 1.25 Ga) dykes from central North Greenland. Geophysical Journal of the Royal Astronomical Society 91, 597–611.

Biggin, A.J., van Hinsbergen, D.J.J., Langereis, C.G., Straathof, G.B., Deenen, M.H.L., 2008. Geomagnetic secular variation in the Cretaceous Normal Superchron and in the Jurassic. Physics of the Earth and Planetary Interiors 169, 3–19.

Buchan, K.L., Halls, H.C., 1990. Palaeomagnetism of proterozoic mafic dyke swarms of the Canadian Shield. In: Parker, A.J., Rickwood, P.C., Tucker, D.H. (Eds.), Mafic Dykes and Emplacement Mechanisms. A.A. Balkema, Rotterdam, pp. 209–230.

Buchan, K.L., Mertanen, S., Park, R.G., Pesonen, L.J., Elming, S.Å., Abrahamsen, N., Bylund, G., 2000. Comparing the drift of Laurentia and Baltica in the Proterozoic: the importance of key palaeomagnetic poles. Tectonophysics 319, 167–198.

Condie, K.C., 2002. Breakup of a Paleoproterozoic supercontinent. Gondwana Research 5, 41–43.

David, J.D., 1983. Determination of domain structure in igneous rocks by alternating field and other methods. Earth and Planetary Science Letters 63, 353–367.

Day, R., Fuller, M., Schmidt, V.A., 1977. Hysteresis properties of titanomagnetites: grain-size and compositional dependence. Physics of the Earth and Planetary Interiors 13, 260–267.
Dong, S., Zhang, Y., Long, C., Yang, Z., Ji, Q., Wang, T., Hu, J., Chen, X., 2007. Jurassic tectonic revolution in China and new interpretation of the Yanshan movement. Acta Geologica Sinica 81, 1449–1461.
Dunlop, D.J., 2002. Theory and application of the Day plot (Mrs/Ms versus Hcr/Hc). 1. Theoretical curves and tests using titanomagnetite data. Journal of Geophysical Research 107, 2056.
Dunlop, D.J., Yu, Y.J., 2004. Intensity and polarity of the geomagnetic field during Precambrian time. In: Channell, J.E.T., Kent, D.V., Lowrie, W., Meert, J.G. (Eds.), Timescales of the Paleomagnetic Field. AGU Monograph. American Geophysical Union, Washington, DC, pp. 85–100.
Ernst, R.E., Buchan, K.L., Hamilton, M.A., Okrugin, A.V., Tomshin, M.D., 2000. Integrated paleomagnetism and U–Pb geochronology of mafic dikes of the Eastern Anabar Shield Region, Siberia: implications for mesoproterozoic paleolatitude of Siberia and comparison with Laurentia. The Journal of Geology 108, 381–401.
Ernst, R.E., Wingate, M.T.D., Buchan, K.L., Li, Z.X., 2008. Global record of 1600–700 Ma large igneous provinces (LIPs): implications for the reconstruction of the proposed Nuna (Columbia) and Rodinia supercontinents. Precambrian Research 160, 159–178.
Evans, D.A.D., Pisarevsky, S.A., 2008. Plate tectonics on early Earth? Weighing the paleomagnetic evidence. Geological Society of America Special Papers 440, 249–263.
Evans, D.A.D., Mitchell, R.N., 2011. Assembly and breakup of the core of Paleoproterozoic–Mesoproterozoic supercontinent Nuna. Geology 39, 443–446.
Fisher, R.A., 1953. Dispersion on a sphere. Proceedings of the Royal Society London, Series A 217, 295–305.
Gao, L.Z., Zhang, C.H., Shi, X.Y., Song, B., Wang, Z.Q., Liu, Y.M., 2008. Mesoproterozoic age for Xiamaling Formation in North China plate indicated by zircon SHRIMP dating. Chinese Science Bulletin 53, 2665–2671.
Guo, J.H., O'Brien, P.J., Zhai, M., 2002. High-pressure granulites in the Sanggan area, North China Craton: metamorphic evolution, P–T paths and geotectonic significance. Journal of Metamorphic Geology 20, 741–756.
Halls, H.C., Li, J., Davis, D., Hou, G., Zhang, B., Qian, X., 2000. A precisely dated Proterozoic palaeomagnetic pole from the North China Craton, and its relevance to palaeocontinental reconstruction. Geophysical Journal International 143, 185–203.
Hawkesworth, C., Cawood, P., Kemp, T., Storey, C., Dhuime, B., 2009. Geochemistry: a matter of preservation. Science 323, 49–50.
He, Y.H., Zhao, G.C., Sun, M., 2008. Geochemistry of the Xiyanghe volcanics at the southern margin of the North China Craton: petrogenesis and tectonic implications. Lithos 102, 158–178.
He, Y.H., Zhao, G.C., Sun, M., Xia, X.P., 2009. SHRIMP and LA-ICP-MS zircon geochronology of the Xiong'er volcanic rocks: implications for the Paleo-Mesoproterozoic evolution of the southern margin of the North China Craton. Precambrian Research 168, 213–222.
He, Y.H., Zhao, G.C., Sun, M., Han, Y., 2010a. Petrogenesis and tectonic setting of volcanic rocks in the Xiaoshan and Waifangshan areas along the southern margin of the North China Craton: constraints from bulk-rock geochemistry and Sr–Nd isotopic composition. Lithos 114, 186–199.
He, Y.H., Zhao, G.C., Sun, M., 2010b. Geochemical and isotopic study of the Xiong'er volcanic rocks at the southern margin of the North China Craton: Petrogenesis and tectonic implications. The Journal of Geology 118, 417–433.
Hou, G., Santosh, M., Qian, X., Lister, G.S., Li, J., 2008a. Tectonic constraints on 1.3–1.2 Ga final breakup of Columbia supercontinent from a giant radiating dyke swarm. Gondwana Research 14, 561–566.
Hou, G., Santosh, M., Qian, X., Lister, G.S., Li, J., 2008b. Configuration of the Late Paleoproterozoic supercontinent Columbia: insights from radiating mafic dyke swarms. Gondwana Research 14, 395–409.
Huang, B.C., Yang, Z.Y., Otofuji, Y., Zhu, R.X., 1999. Early Paleozoic paleomagnetic poles from the western part of the North China Block and their implications. Tectonophysics 308 (3), 377–402.
Huang, B.C., Zhou, Y.X., Zhu, R.X., 2008. Discussions on Phanerozoic evolution and formation of continental China, based on paleomagnetic studies. Earth Science Frontiers 15 (3), 348–359 (in Chinese).
Irving, E., Baker, J., Hamilton, M., Wynne, P.J., 2004. Early Proterozoic geomagnetic field in western Laurentia: implications for paleolatitudes: local rotations and stratigraphy. Precambrian Research 129, 251–270.
Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of palaeomagnetic data. Geophysical Journal of the Royal Astronomical Society 62, 699–718.
Li, H.K., Lu, S.N., Li, H.M., Sun, L.X., Xiang, Z.Q., Geng, J.Z., Zhou, H.Y., 2009. Zircon and beddeleyite U–Pb precision dating of basic rock sills intruding Xiamaling Formation, North China. Geological Bulletin of China 28, 1396–1404.
Li, H., Zhu, S.X., Xiang, Z.Q., Su, W.B., Lu, S.N., Zhou, H.Y., Geng, J.Z., Li, S., Yang, F.J., 2010. Zircon U–Pb dating on tuff bed from Gaoyuzhuang Formation in Yanqing, Beijing: further constraints on the new subdivision of the Mesoproterozoic stratigraphy in the northern North China. Acta Petrologica Sinica 26, 2131–2140.
Li, S.Z., Zhao, G.C., Sun, M., Han, Z., Luo, Y., Hao, D., Xia, X.P., 2005. Deformation history of the Paleoproterozoic Liaohe assemblages in the Eastern Block of the North China Craton. Journal of Asian Earth Sciences 24, 659–674.
Li, S.Z., Zhao, G.C., Sun, M., Han, Z.Z., Zhao, G.T., Hao, D.F., 2006. Are the South and North Liaohe Groups of North China Craton different exotic terranes? Nd isotope constraints. Gondwana Research 9, 198–208.
Li, S.Z., Zhao, G.C., 2007. SHRIMP U–Pb zircon geochronology of the Liaoji granitoids: constraints on the evolution of the Paleoproterozoic Jiao–Liao–Ji belt in the Eastern Block of the North China Craton. Precambrian Res 158, 1–16.
Li, S.Z., Zhao, G.C., Zhang, J., Sun, M., Zhang, G.W., Luo, D., 2010. Deformational history of the Hengshan–Wutai–Fuping belt: implications for the evolution of the Trans-North China Orogen. Gondwana Research 18, 611–631.
Li, S.Z., Zhao, G.C., Santosh, M., Liu, X., Dai, L.M., Suo, Y.H., Tam, P.Y., Song, M.C., Wang, P.C., 2012. Paleoproterozoic structural evolution of the southern segment of the Jiao–Li Ji Belt, North China Craton. Precambrian Research 200–203, 59–73.
Li, Z.X., Zhang, L., Powell, C.M., 1996. Positions of the East Asian cratons in the Neoproterozoic supercontinent Rodinia. Australian Journal of Earth Sciences 43, 593–604.
Linder, J., Gilder, S.A., 2012. Latitude dependency of the geomagnetic secular variation S parameter: a mathematical artifact. Geophysical Research Letters 39, L02308.
Lu, S., Zhao, G., Wang, H., Hao, G., 2008. Precambrian metamorphic basement and sedimentary cover of the North China Craton: a review. Precambrian Research 160, 77–93.
Lu, X.P., Wu, F.Y., Guo, J.H., Wilde, S.A., Yang, J.H., Liu, X.M., Zhang, X.O., 2006. Zircon U–Pb geochronological constraints on the Paleoproterozoic crustal evolution of the Eastern block in the North China Craton. Precambrain Research 146, 138–164.
Lubnina, N.V., 2009. The East European Craton in the Mesoproterozoic: new key paleomagnetic poles. Doklady Earth Sciences 428, 1174–1178.
Lubnina, N.V., Mertanen, S., Söderlund, U., Bogdanova, S., Vasilieva, T.I., Frank-Kamenetsky, D., 2010. A new key pole for the East European Craton at 1452 Ma: palaeomagnetic and geochronological constraints from mafic rocks in the Lake Ladoga region (Russian Karelia). Precambrian Research 183, 442–462.
Marcussen, C., Abrahamsen, N., 1983. Palaeomagnetism of the proterozoic zig–zag dal Basalt and the Midsommersø Dolerites, eastern North Greenland. Geophysical Journal of the Royal Astronomical Society 73, 367–387.
McFadden, P.L., 1990. A new fold test for palaeomagnetic studies. Geophysical Journal International 103, 163–169.
Meert, J.G., Stuckey, W., 2002. Revisiting the paleomagnetism of the 1.476 Ga St. Francois Mountains igneous province, Missouri. Tectonics 21, 1007.
Meert, J.G., Pandit, M.K., Pradhan, V.R., Kamenov, G., 2011. Preliminary report on the paleomagnetism of 1.88 Ga dykes from the Bastar and Dharwar cratons, Peninsular India. Gondwana Research 20, 335–343.
Meng, Q.-R., Wei, H.-H., Qu, Y.-Q., Ma, S.-X., 2011. Stratigraphic and sedimentary records of the rift to drift evolution of the northern North China Craton at the Paleo- to Mesoproterozoic transition. Gondwana Research 20, 205–218.
Pei, J., Yang, Z., Zhao, Y., 2006. A Mesoproterozoic paleomagnetic pole from the Yangzhuang Formation, North China and its tectonics implications. Precambrian Research 151, 1–13.
Peng, P., Zhai, M., Zhang, H., Guo, J., 2005. Geochronological constraints on the Paleoproterozoic evolution of the North China Craton: SHRIMP zircon ages of different types of mafic dikes. International Geology Review 47, 492–508.
Peng, P., 2010. Reconstruction and interpretation of giant mafic dyke swarms: a case study of 1.78 Ga magmatism in the North China Craton. Geological Society London, Special Publications 338, 163–178.
Piper, J.D.A., 2010. Protopangaea: palaeomagnetic definition of Earth's oldest (mid-Archaean-Palaeoproterozoic) supercontinent. Journal of Geodynamics 50, 154–165.
Piper, J.D.A., Zhang, J.S., Huang, B.C., Roberts, A.P., 2011. Palaeomagnetism of Precambrian dyke swarms in the North China Shield: the ~1.8 Ga LIP event and crustal consolidation in late Palaeoproterozoic times. Journal of Asian Earth Sciences 41 (6), 504–524.
Pisarevsky, S.A., Sokolov, S.J., 2001. The magnetostratigraphy and a 1780 Ma palaeomagnetic pole from the red sandstones of the Vazhinka River section, Karelia, Russia. Geophysical Journal International 146, 531–538.
Pisarevsky, S.A., Natapov, L.M., 2003. Siberia and Rodinia. Tectonophysics 375, 221–245.
Rogers, J.J.W., Santosh, M., 2002. Configuration of Columbia, a Mesoproterozoic Supercontinent. Gondwana Research 5, 5–22.
Salminen, J., Pesonen, L.J., 2007. Paleomagnetic and rock magnetic study of the Mesoproterozoic sill, Valaam island, Russian Karelia. Precambrian Research 159, 212–230.
Su, W., Zhang, S., Huff, W.D., Li, H., Ettensohn, F.R., Chen, X., Yang, H., Han, Y., Song, B., Santosh, M., 2008. SHRIMP U–Pb ages of K-bentonite beds in the Xiamaling Formation: implications for revised subdivision of the Meso- to Neoproterozoic history of the North China Craton. Gondwana Research 14, 543–553.
Su, W., Li, H., Huff, W.D., Ettensohn, F.R., Zhang, S., Zhou, H., Wan, Y., 2010. SHRIMP U–Pb dating for a K-bentonite bed in the Tieling Formation, North China. Chinese Science Bulletin 55, 3312–3323.
Su, W., Li, H., Xu, L., Jia, S., Geng, J., Zhou, H., Wang, Z., Pu, H., 2012. Luoyu and Ruyang Group at the south margin of the North China Craton (NCC) should belong in the Mesoproterozoic Changchengian system: direct constraints from the LA-MC-ICPMS U–Pb age of the tuffite in the Luoyukou Formation, Ruzhou, Henan, China. Geological Survey and Research 35 (2), 96–108 (in Chinese).
Tam, P.Y., Zhao, G.C., Liu, F.L., Zhou, X.W., Sun, M., Li, S.Z., 2011. Timing of metamorphism in the Paleoproterozoic Jiao–Liao–Ji Belt: new SHRIMP U–Pb zircon dating of granulites, gneisses and marbles of the Jiaobei massif in the North China Craton. Gondwana Research 19, 150–162.
Tam, P.Y., Zhao, G.C., Zhou, X.W., Sun, M., Guo, J.H., Li, S.Z., Yin, C.Q., Wu, M.L., He, Y.H., 2012. Metamorphic P–T path and implications of high-pressure pelitic

granulites from the Jiaobei massif in the Jiao–Liao–Ji Belt, North China Craton. Gondwana Research 22, 104–117.
Upton, B.G.J., Rämö, O.T., Heaman, L.M., Blichert-Toft, J., Kalsbeek, F., Barry, T.L., Jepsen, H.F., 2005. The Mesoproterozoic Zig–Zag Dal basalts and associated intrusions of eastern North Greenland: mantle plume–lithosphere interaction. Contributions to Mineralogy and Petrology 149, 40–56.
Vandamme, D., 1994. A new method to determine paleosecular variation. Physics of the Earth and Planetary Interiors 85, 131–142.
Wang, J., Wu, Y.B., Gao, S., Peng, M., Liu, X.C., Zhao, L.S., Zhou, L., Hu, Z.C., Gong, H.J., Liu, Y.S., 2010. Zircon U–Pb and trace element data from rocks of the Huai'an Complex: new insights into the late Paleoproterozoic collision between the Eastern and Western Blocks of the North China Craton. Precambrian Research 178, 59–71.
Wang, S.S., Sang, H.Q., Qiu, J., Chen, M.E., Li, M.R., 1995. The forming ages of Yangzhuang and Wumishan formations in Jixian section, Northern China. Science Geologica Sinica 30, 166–176.
Watson, G.S., Enkin, R.J., 1993. The fold test in paleomagnetism as a parameter estimation problem. Geophysical Research Letters 20, 2135–2137.
Wilde, S.A., Zhao, G., Sun, M., 2002. Development of the North China Craton during the late Archaean and its final amalgamation at 1.8 Ga: some speculations on its position within a global Palaeoproterozoic Supercontinent. Gondwana Research 5, 85–94.
Wingate, M.T.D., Pisarevsky, S.A., Gladkochub, D.P., Donskaya, T.V., Konstantinov, K.M., Mazukabzov, A.M., Stanevich, A.M., 2009. Geochronology and paleomagnetism of mafic igneous rocks in the Olenek Uplift, northern Siberia: implications for Mesoproterozoic supercontinents and paleogeography. Precambrian Research 170, 256–266.
Wu, H.C., Zhang, S.H., Li, Z.X., Li, H.Y., Dong, J., 2005. New paleomagnetic results from the Yangzhuang Formation of the Jixian System, North China, and tectonic implications. Chinese Science Bulletin 50 (14), 1483–1489.
Xia, X.P., Sun, M., Zhao, G.C., Luo, Y., 2006a. LA-ICP-MS U–Pb geochronology of detrital zircons from the Jining Complex, North China Craton and its tectonic significance. Precambrian Research 144, 199–212.
Xia, X.P., Sun, M., Zhao, G.C., Wu, F.Y., Xu, P., Zhang, J.H., Luo, Y., 2006b. U–Pb and Hf isotopic study of detrital zircons from the Wulashan khondalites: constraints on the evolution of the Ordos Terrane, Western Block of the North China Craton. Earth and Planetary Science Letters 241, 581–593.
Xia, X.P., Sun, M., Zhao, G.C., Wu, F.Y., Xu, P., Zhang, J.S., 2008. Paleoproterozoic crustal growth events in the Western Block of the North China Craton: evidence from detrital zircon Hf and whole rock Sr–Nd isotopes of the khondalites in the Jining Complex. American Journal of Science 308, 304–327.
Xia, X.P., Sun, M., Zhao, G.C., Wu, F.Y., Xu, P., Zhang, J.S., 2009. Detrital zircon U–Pb age and Hf isotope study of the khondalite in Trans-North China Orogen and its tectonic significance. Geological Magazine 146, 701–716.
Yang, K.-F., Fan, H.-R., Santosh, M., Hu, F.-F., Wang, K.-Y., 2011. Mesoproterozoic mafic and carbonatitic dykes from the northern margin of the North China Craton: implications for the final breakup of Columbia supercontinent. Tectonophysics 498, 1–10.
Yang, Z.Y., Otofuji, Y., Sun, Z.M., Huang, B.C., 2002. Magnetostratigraphic constraints on the Gondwanan origin of North China: Cambrian/Ordovician boundary results. Geophysical Journal International 151, 1–10.
Yin, C.Q., Zhao, G.C., Sun, M., Xia, X.P., Wei, C.J., Zhou, X.W., Leung, W.H., 2009. LA-ICP-MS U–Pb zircon ages of the Qianlishan complex: constraints on the evolution of the Khondalite Belt in the Western Block of the North China Craton. Precambrian Research 174, 78–94.
Yin, C.Q., Zhao, G.C., Guo, J.H., Sun, M., Xia, X.P., Zhou, X.W., Liu, C.H., 2011. U–Pb and Hf iso-topic study of zircons of the Helanshan Complex: constrains on the evolution of the Khodalite Belt in theWestern Block of the North China Craton. Lithos 122, 25–38.
Zhai, M.G., Meng, Q.R., Liu, J.M., Hou, Q.L., Hu, S.B., Li, Z., Zhang, H.F., Liu, W., Shao, J., Zhu, R.X., 2004. Geological features of Mesozoic tectonic regime inversion in Eastern North China and implication for geodynamics. Earth Science Frontiers 11, 285–297.
Zhang, J., Zhao, G.C., Li, S.Z., Sun, M., Wilde, S.A., Liu, S.W., Yin, C.Q., 2009. Deformational history of the Fuping Complex and new U–Th–Pb geochronological constraints: implications for the tectonic evolution of the Trans-North China Orogen. Journal of Structural Geology 31, 177–193.
Zhang, S.-H., Zhao, Y., Yang, Z.-Y., He, Z.-F., Wu, H., 2009. The 1.35 Ga diabase sills from the northern North China Craton: implications for breakup of the Columbia (Nuna) supercontinent. Earth and Planetary Science Letters 288, 588–600.
Zhang, S., Li, Z., Wu, H., 2006. New Precambrian palaeomagnetic constraints on the position of the North China Block in Rodinia. Precambrian Research 144, 213–238.
Zhang, S., Li, Z.-X., Evans, D.A.D., Wu, H., Li, H., Dong, J., 2012. Pre-Rodinia supercontinent Nuna shaping up: a global synthesis with new paleomagnetic results from North China. Earth and Planetary Science Letters 353-354, 145–155.
Zhao, G., Cawood, P.A., Wilde, S.A., Sun, M., Lu, L., 2000. Metamorphism of basement rocks in the Central Zone of the North China Craton: implications for Paleoproterozoic tectonic evolution. Precambrian Research 103, 55–88.
Zhao, G., Cawood, P.A., Wilde, S.A., Lu, L., 2001. High-pressure granulites (retrograded eclogites) from the Hengshan Complex, North China Craton: petrology and tectonic implications. Journal of Petrology 42, 1141–1170.
Zhao, G., Sun, M., Wilde, S.A., Li, S., 2003a. Assembly, accretion and breakup of the Paleo-Mesoproterozoic Columbia Supercontinent: records in the North China Craton. Gondwana Research 6, 417–434.
Zhao, G., Sun, M., Wilde, S.A., 2003b. Correlations between the Eastern Block of the North China Craton and the South Indian Block of the Indian Shield: an Archaean to Palaeoproterozoic link. Precambrian Research 122, 201–233.
Zhao, G., Sun, M., Wilde, S.A., Li, S., 2004. A Paleo-Mesoproterozoic supercontinent: assembly, growth and breakup. Earth Science Reviews 67, 91–123.
Zhao, G.C., Sun, M., Wilde, S.A., Li, S.Z., 2005. Late Archean to Paleoproterozoic evolution of the North China Craton: key issues revisited. Precambrian Research 136, 177–202.
Zhao, G.C., He, Y.H., Sun, M., 2009. The Xiong'er volcanic belt at the southern margin of the North China Craton: petrographic and geochemical evidence for its outboard position in the Paleo-Mesoproterozoic Columbia Supercontinent. Gondwana Research 16, 170–181.
Zhao, G.C., Wilde, S.A., Guo, J.H., Cawood, P.A., Sun, M., Li, X.P., 2010. Single zircon grains record two Paleoproterozoic collisional events in the North China Craton. Precambrian Research 177, 266–276.
Zhao, G., Li, S., Sun, M., Wilde, S.A., 2011. Assembly, accretion, and breakup of the Palaeo-Mesoproterozoic Columbia supercontinent: record in the North China Craton revisited. International Geology Review 53, 1331–1356.
Zhao, Y., Xu, G., Zhang, S.-H., Yang, Z.-Y, Zhang, Y.Q., Hu, J.M., 2004. Yanshanian movement and conversion oftectonic regimes in East Asia. Earth Science Frontiers 11, 319–328.
Zhou, X.W., Zhao, G.C., Wei, C., Geng, Y.S., Sun, M., 2008. EPMA U–Pb monazite and Shrimp U–Pb zircon Geochronology of high-pressure pelitic granulites in the Jiaobei massif of the North China Craton. American Journal of Science 308, 328–350.
Zijderveld, J.D.A., 1967. A.C. demagnetization of rocks: analysis of results. In: Collinson, D.W., Creer, K.M., Runcorn, S.K. (Eds.), Methods on Paleomagnetism. Elsevier, Amsterdam, pp. 254–286.